



Exploring the World of Science

B04

Geffen Academy at UCLA

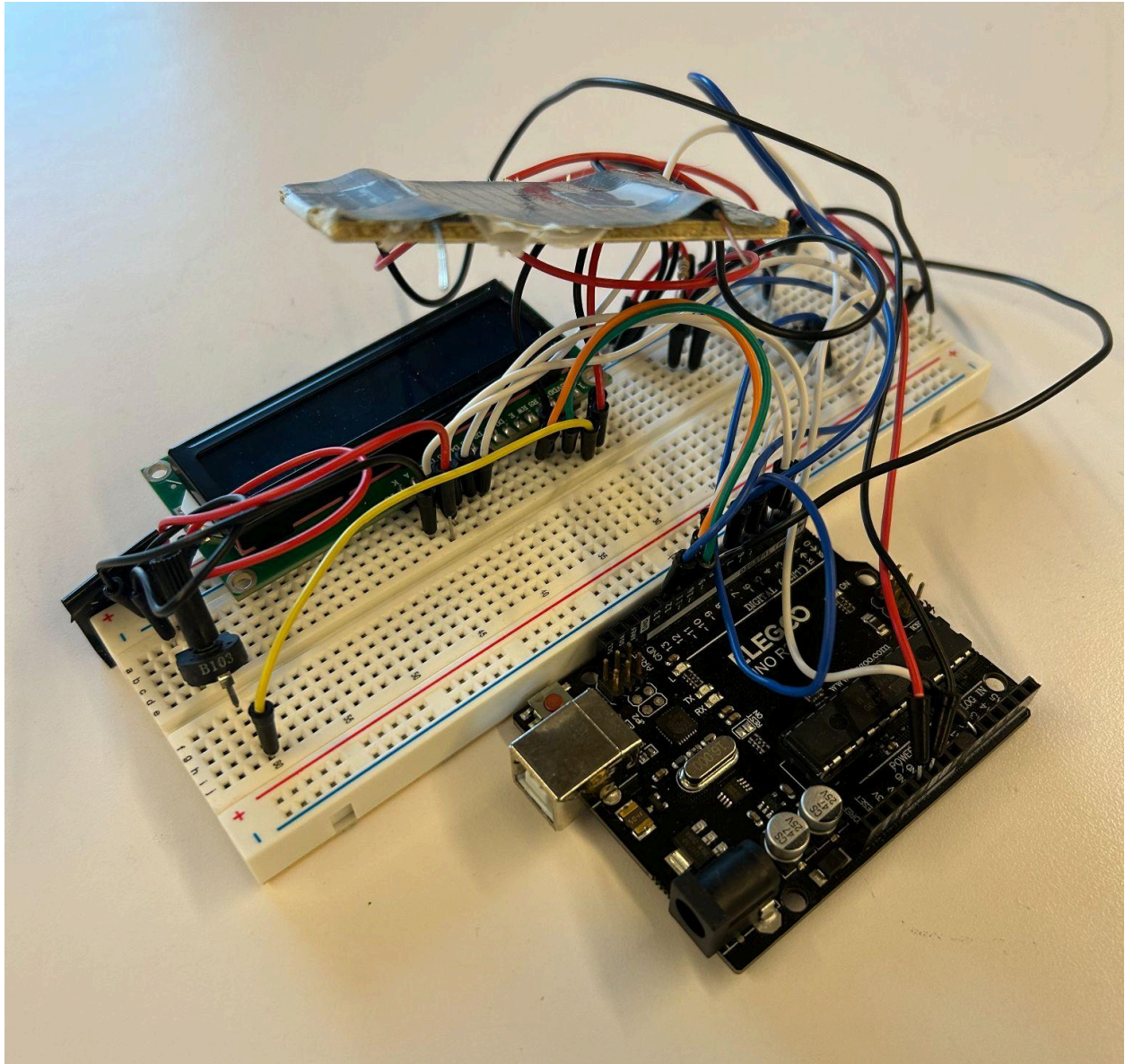
Water Quality

Design log

Antelope Valley College - Division B

February 14th, 2026

Salinometer



Build:

Materials:

- 1x Elegoo Arduino Uno R3
- Jumper cables
- 1x DS18B20 Waterproof Temperature Sensor
- 3x Resistor: 1x 10k Ohm, 1x 1kOhm, 1x 4.7kOhm
- 1x Button
- 1x LCD 1602
- 1x Breadboard

Instructions:

1. Use a resistor to connect two columns of the breadboard
2. Use a jumper cable to connect an analog pin to one end of the resistor
3. Place the electrode on the same column as the analog pin.
4. Connect a digital pin to the end of the resistor without the analog pin.
5. Connect another digital pin to an electrode
6. Connect a digital pin to a column of the breadboard
7. Attach one side of the button to that column. Connect the end diagonal from the end with the digital pin to ground.
8. Attach the end across from the digital pin with a 10k ohm resistor to power.
9. Attach the LCD to the breadboard.
10. Attach VSS and VDD to ground and power, respectively.
11. Attach V0 to GND
12. Attach RW to GND
13. Attach RS and D4-7 to digital pins
14. Attach A to power and K to ground

Programing:

```
#include <Wire.h>
#include <LiquidCrystal.h>

LiquidCrystal lcd(11,10,2,3,4,9); //Pins

// --- Settings ---
int analogPin = 0;
int buttonPin = 5;
int dtime = 500; // Delay time
int samples = 100; //Measurement sample size
float Vin = 5.0;
float R1 = 1000.0;

// --- Variables ---
bool isRunning = false; // false = Stopped, true = Measuring
int buttonState; // Current reading from the input pin
int lastButtonState = HIGH; // Previous reading (default HIGH for PULLUP)

unsigned long lastDebounceTime = 0;
unsigned long debounceDelay = 50; // 50ms to filter noise

void setup(){
  pinMode(8, OUTPUT);
  pinMode(7, OUTPUT);
  pinMode(12, INPUT);
```

```

pinMode(buttonPin, INPUT_PULLUP);
Serial.begin(9600);
lcd.begin(16, 2);
lcd.clear();
lcd.print("Ready");
}

void loop(){
  // 1. READ BUTTON
  int reading = digitalRead(buttonPin);

  // If the switch changed, due to noise or pressing:
  if (reading != lastButtonState) {
    lastDebounceTime = millis();
  }

  // Only check if enough time has passed (Debounce)
  if ((millis() - lastDebounceTime) > debounceDelay) {

    // If the button state has changed:
    if (reading != buttonState) {
      buttonState = reading;

      // If the new state is LOW, it means the button was just pressed
      if (buttonState == LOW) {
        isRunning = !isRunning;

        // Update LCD immediately upon press
        lcd.clear();
        if (isRunning) {
          lcd.print("Measuring...");
          Serial.println("STARTED");
        } else {
          lcd.print("Stopped");
          Serial.println("STOPPED");
          delay(500);
          lcd.clear();
          lcd.print("Ready");
        }
      }
    }
  }
}

```

```

    // Save the reading. Next time through the loop, it'll be the
lastButtonState:
    lastButtonState = reading;

// 2. MEASURE (Only if isRunning is true)
if (isRunning) {
    float tot = 0;
    float avg = 0;
    int raw = 0;
    float Vout = 0;
    float R2 = 0;
    float buff = 0;

    // Measurement batch
    for (int i = 0; i < samples; i++) {
        digitalWrite(7, HIGH);
        digitalWrite(8, LOW);
        delayMicroseconds(dtime);
        digitalWrite(7, LOW);
        digitalWrite(8, HIGH);
        delayMicroseconds(dtime);

        raw = analogRead(analogPin);

        if(raw > 0){
            buff = raw * Vin;
            Vout = (buff)/1024.0;
            // Prevent division by zero
            if(Vout > 0.01) {
                buff = (Vin/Vout) - 1;
                R2 = R1 * buff;
                tot = tot + R2;
            }
        }
    }

    // Calculate and print
    if (tot > 0) {
        avg = tot / samples;
        // Salt percent formula
        float calcVal = ((-151.00/860.00)*avg) + (514481.00/10750.00) + 0.8;

```

```

    lcd.setCursor(0, 0);
    lcd.print("R: ");
    lcd.print((int)avg);

    lcd.setCursor(0, 1);
    lcd.print("Val: ");
    lcd.print(calcVal);
  }
}
}

```

Testing:

To calibrate the salinometer to display percent salt instead of resistance, I used a real water quality tester to find the salt percentage, then measured the resistance. I then used several data points to then graph the line of best fit. This is how I came up with the formula $\text{Salt\%} = ((-151.00/860.00) * \text{res}) + (514481.00/10750.00) + 0.8$.

Salt Percent	Resistance Value
4.08	47.66
8.16	48.52
12.24	49.55

About this item

<https://a.co/d/0c9FXtGB>

ORAPXI pH and Salt Meter for Saltwater
Pool pH and Salinity Digital Tester

Test pool salt level- Salt range:0~9990ppm/10~100ppt;
Essential for keeping your pool salt water concentrate at the correct level if you are using a salt water chlorine generator.
Check the salt concentration of saltwater pool and hot tub that use salt chlorine generator; Help the salt chlorinator work more efficiently in the appropriate salinity value.



Water
Quality
Tester